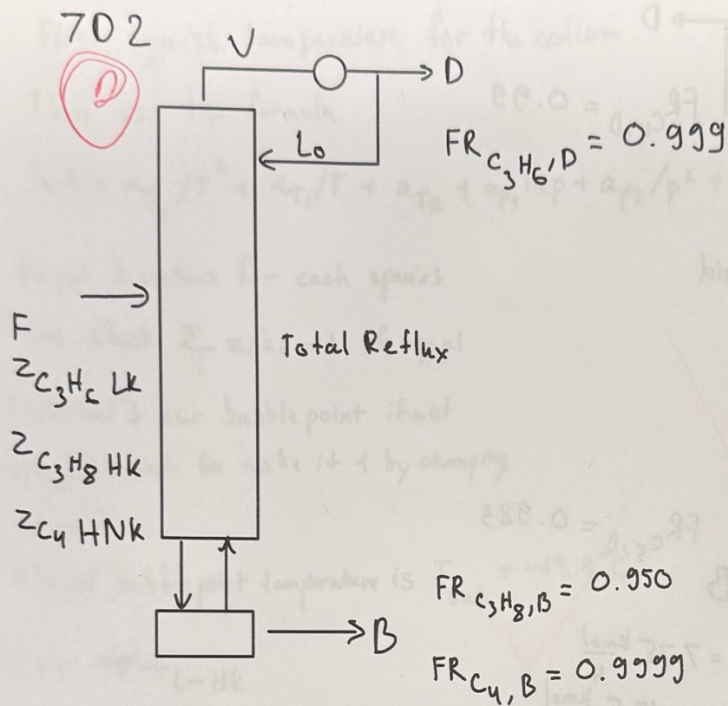


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using Fenske eq

At 34°C

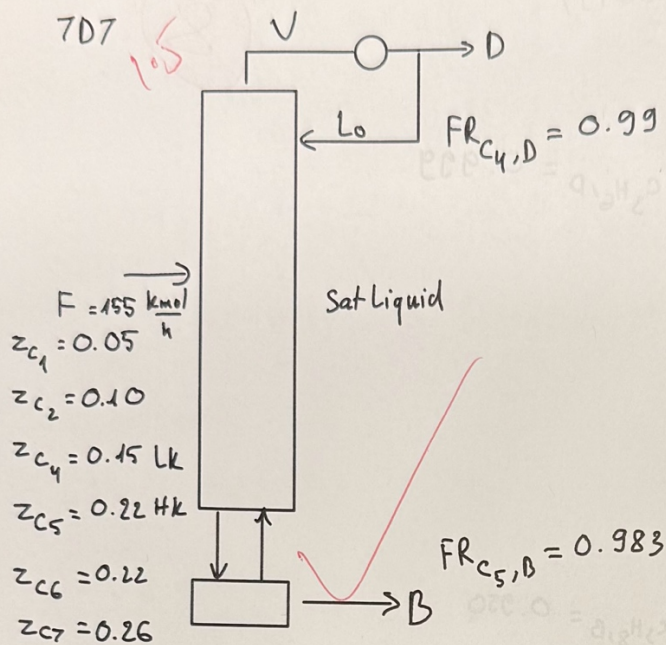
$$\alpha_{LK-HK} = \frac{k_{C_3H_6}}{k_{C_3H_8}} = \frac{1.00}{0.89} = 1.1236$$

At 40°C

$$\alpha_{LK-HK} = \frac{1.13}{1.000} = 1.130$$

$$\text{take } \alpha_{LK-HK} = \sqrt{1.1236 \times 1.130} = 1.12697$$

$$N_{\min} = \frac{\ln \left[\frac{0.999 \times 0.950}{(1-0.999)(1-0.950)} \right]}{\ln(1.12697)} = 82.41 \text{ stages}$$



a) $Dx_{c1,D} = Fz_{c1} = 155 \times 0.05 = 7.75 \frac{\text{kmol}}{\text{h}}$

$Dx_{c2,D} = Fz_{c2} = 155 \times 0.10 = 15.5 \frac{\text{kmol}}{\text{h}}$

$Dx_{c4,D} = FR_{c4,D} \cdot Fz_{c4} = 0.99 \cdot 155 \times 0.15 = 23.0175 \frac{\text{kmol}}{\text{h}}$

$Dx_{c5,D} = FR_{c5,D} \cdot Fz_{c5} = (1 - FR_{c5,B}) \cdot Fz_{c5}$
 $= (1 - 0.983) \cdot 155 \times 0.22 = 0.5797 \frac{\text{kmol}}{\text{h}}$

$Dx_{c6,D} = 0 \frac{\text{kmol}}{\text{h}}$

$Dx_{c7,D} = 0 \frac{\text{kmol}}{\text{h}}$

B=77

$\Rightarrow D = \sum_i Dx_{i,D} = 7.75 + 15.5 + 23.0175 + 0.5797$
 $= 46.8472 \frac{\text{kmol}}{\text{h}}$

$x_{c1,D} = 0.1654 \quad x_{c5,D} = 0.01237$

$x_{c2,D} = 0.3309 \quad x_{c6,D} = 0$

$x_{c4,D} = 0.4913 \quad x_{c7,D} = 0$

b) To get Bubble point temperature

First guess temperature for the column

Then use the formula

$$\ln k = \alpha_{T_1}/T^2 + \alpha_{T_2}/T + \alpha_{T_6} + \alpha_{p_1} \ln p + \alpha_{p_2}/p^2 + \alpha_{p_3}/p$$

to get k values for each species

Then check $\sum_i z_i k_i = 1$ if equal

then that's our bubble point if not
use goal seek to make it 1 by changing
temperature

- Feed bubble point temperature is $T_{bub} = -14.8^\circ\text{C}$

Now α_{i-C_5}

$$\alpha_{C_1-C_5} = \frac{k_{C_1}}{k_{C_5}} = 733.551$$

$$\alpha_{C_2-C_5} = \frac{k_{C_2}}{k_{C_5}} = 88.06$$

$$\alpha_{C_4-C_5} = " = 4.097$$

$$\alpha_{C_5-C_5} = " = 1$$

$$\alpha_{C_6-C_5} = " = 0.2646$$

$$\alpha_{C_7-C_5} = " = 0.07288$$

We have already calculate the composition of the distillate

so now $\sum_i x_{i,D} k_i = 1$

$T_{bub,D} = -78.05^\circ$ too cold cryogenics require expensive

$$c) N_{\min} = \frac{\ln\left(\frac{0.99 \times 0.983}{(1-0.99)(1-0.983)}\right)}{\ln 4.097} = 6.135 \text{ stages}$$

d) Sat liquid $q = 1$

$$\Rightarrow \Delta V_{\text{feed}} = F(1-q) = F(1-1) = 0$$

$$\Delta V_{\text{feed}} = \sum_i \frac{\alpha_{i-\text{ref}} F z_i}{\alpha_{i-\text{ref}} - \phi}$$

$$0 = \frac{\alpha_{c_1-c_5} F z_{c_1}}{\alpha_{c_1-c_5} - \phi} + \frac{\alpha_{c_2-c_5} F z_{c_2}}{\alpha_{c_2-c_5} - \phi} + \frac{\alpha_{c_4-c_5} F z_{c_4}}{\alpha_{c_4-c_5} - \phi} + \frac{\alpha_{c_5-c_5} F z_{c_5}}{\alpha_{c_5-c_5} - \phi} + \frac{\alpha_{c_6-c_5} F z_{c_6}}{\alpha_{c_6-c_5} - \phi} + \frac{\alpha_{c_7-c_5} F z_{c_7}}{\alpha_{c_7-c_5} - \phi}$$

Solve for $\phi = 1.634$

$$V_{\min} = \sum_i \frac{\alpha_{i-\text{ref}} (D x_{i,D})}{\alpha_{i-\text{ref}} - \phi}$$

$$= \frac{\alpha_{c_1-c_5} (D x_{c_1,D})}{\alpha_{c_1-c_5} - \phi} + \frac{\alpha_{c_2-c_5} (D x_{c_2,D})}{\alpha_{c_2-c_5} - \phi} + \frac{\alpha_{c_4-c_5} (D x_{c_4,D})}{\alpha_{c_4-c_5} - \phi} + \frac{\alpha_{c_5-c_5} (D x_{c_5,D})}{\alpha_{c_5-c_5} - \phi} +$$

$$\frac{\alpha_{c_6-c_5} (D x_{c_6,D})}{\alpha_{c_6-c_5} - \phi} + \frac{\alpha_{c_7-c_5} (D x_{c_7,D})}{\alpha_{c_7-c_5} - \phi} \quad \text{no } D x_{c_6,D} \text{ and } D x_{c_7,D}$$

$$V_{\min} = 60.94 \frac{\text{kmol}}{\text{h}}$$

$$L_{\min} = V_{\min} - D = 60.94 - 46.847 = 14.093 \frac{\text{kmol}}{\text{h}}$$

$$\left(\frac{L}{D}\right)_{\min} = \frac{14.093}{46.847} = 0.300676$$

$$e) L/D = M \times (L/D)_{\min}$$

$$M = 1.04 \rightarrow L/D = 1.04 \times 0.3 = 0.312$$

$$\frac{L/D - (L/D)_{\min}}{L/D + 1} = 0.009$$

$$\frac{3}{328}$$

$$\frac{N - N_{\min}}{N + 1} = 1 - 18.5715 \times$$

$$\frac{N - 6.135}{N + 1} = 1 - 18.5715 \times 0.009$$

$$\Rightarrow N = 22 \text{ stages}$$

$$M = 1.10$$

$$L/D = 1.10 \times 0.3 = 0.33$$

$$\frac{L/D - (L/D)_{\min}}{L/D + 1} = 0.0226$$

$$N = 19.63 \text{ stages}$$

$$M = 2$$

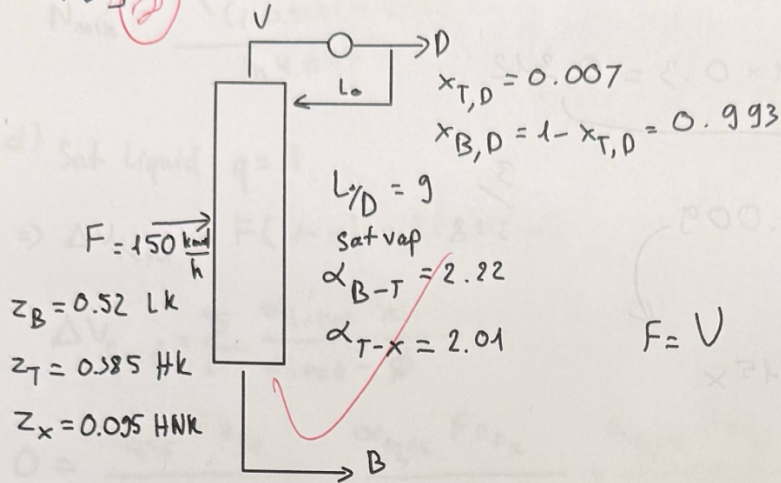
$$L/D = 2 \times 0.3 = 0.6$$

$$\frac{L/D - (L/D)_{\min}}{L/D + 1} = 0.1875$$

$$N = 11.96$$

f) Alternative should be found due to the low temperature require!!

70 g



$$F = V$$

Bottom

$$B x_{x,B} = F z_x = 150 \times 0.095 = 14.25 \frac{\text{kmol}}{\text{h}}$$

$$B = 150 - 15 = 135 \frac{\text{kmol}}{\text{h}}$$

$$x_{D,B} = \frac{F}{B} z_B - \frac{D}{B} x_{B,D}$$

$$= \frac{150}{135} \times 0.52 - \frac{15}{135} \times 0.993 = 0.467$$

$$x_{T,B} = \frac{F}{B} z_T - \frac{D}{B} x_{T,D}$$

$$= \frac{150}{135} \times 0.385 - \frac{15}{135} \times 0.007 = 0.427$$

$$x_{x,B} = 1 - 0.427 - 0.467 = 0.106$$

$$b) N_{min} = \frac{\ln \left(\frac{D x_{B,D} / D x_{T,D}}{B x_{B,B} / B x_{T,B}} \right)}{\ln 2.22} = 6.1$$

$$c) \alpha_{Tol-x_y} = \frac{N_{min} x_{T,D} / x_{x,D}}{x_{T,B} / x_{x,B}}$$

Top

$$V = L_0 + D = 9D + D = 10D$$

$$F = 10D$$

$$\Rightarrow 150 = 10D$$

$$D = 15 \frac{\text{kmol}}{\text{h}}$$

$$\Rightarrow 2.01^{6.1} = \frac{0.007 / x_{x,D}}{0.427 / 0.106}$$

$$\Rightarrow x_{x,D} = 2.45 \times 10^{-5}$$

$$\alpha_{B-T} = \frac{k_B}{k_T}$$

$$\alpha_{T-x} = \frac{k_T}{k_x}$$

$$\alpha_{B-x} = \frac{k_B}{k_x} = \alpha_{B-T} \alpha_{T-x} = 4.4622$$

$$d) \Delta V_{\text{feed}} = F(1-0) = 150 \frac{\text{kmol}}{\text{h}}$$

$$150 = \frac{\alpha_{B-x} F_{z_B}}{\alpha_{B-x} - \phi} + \frac{\alpha_{T-x} F_{z_T}}{\alpha_{T-x} - \phi} + \frac{\alpha_{x-x} F_{z_x}}{\alpha_{x-x} - \phi}$$

$$150 = \frac{4.4622 \times 150 \times 0.52}{4.4622 - \phi} + \frac{2.01 \times 150 \times 0.385}{2.01 - \phi} + \frac{1 \times 150 \times 0.095}{1 - \phi}$$

$$\phi = 3.126$$

$$V_{\min} = \frac{\alpha_{B-x} D_{x_B}}{\alpha_{B-x} - \phi} + \frac{\alpha_{T-x} D_{x_T}}{\alpha_{T-x} - \phi} + \frac{\alpha_{x-x} D_{x_x}}{\alpha_{x-x} - \phi}$$

$$= \frac{4.4622 \times 15 \times 0.993}{4.4622 - \phi} + \frac{2.01 \times 15 \times 0.007}{2.01 - \phi} + \frac{1 \times 15 \times 2.45 \times 10^{-5}}{1 - \phi}$$

$$= 49.55 \frac{\text{kmol}}{\text{h}}$$

$$V = L + D$$

$$V - D = L$$

$$\Rightarrow \left(\frac{L}{D} \right)_{\min} = \left(\frac{V_{\min} - D}{D} \right)_{\min} = \frac{49.55 - 15}{15} = 2.3$$

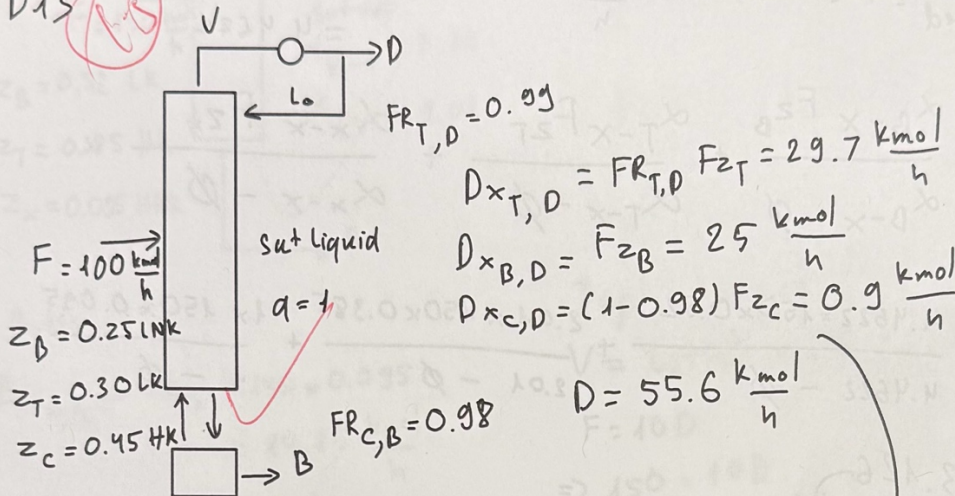
$$e) \frac{L/D - L/D_{\min}}{L/D + 1} = \frac{9 - 2.3}{9 + 1} = 0.67$$

$$\frac{N - 6.1}{N + 1} = 0.545827 - 0.591422 \times 0.67 + \frac{0.002743}{0.67}$$

$$N = 7.4$$

2 stages

D13



$$\Delta V = 0$$

$$\Rightarrow 0 = \frac{\alpha_{BT} F z_B}{\alpha_{BT} - \phi} + \frac{\alpha_{TT} F z_T}{\alpha_{TT} - \phi} + \frac{\alpha_{CT} F z_C}{\alpha_{CT} - \phi}$$

Solve for ϕ

$$\phi = 0.337$$

$$V_{\min} = \frac{\alpha_{BT} \times D x_{B,D}}{\alpha_{BT} - \phi} + \frac{\alpha_{TT} \times D x_{T,D}}{\alpha_{TT} - \phi} + \frac{\alpha_{CT} D x_{C,D}}{\alpha_{CT} - \phi}$$

$$= 72.24 \frac{\text{kmol}}{\text{h}}$$

$$(L/D)_{\min} = \left(\frac{V_{\min}}{D} \right)_{\min} = \frac{72.24 - 55.6}{55.6} = 0.2992$$

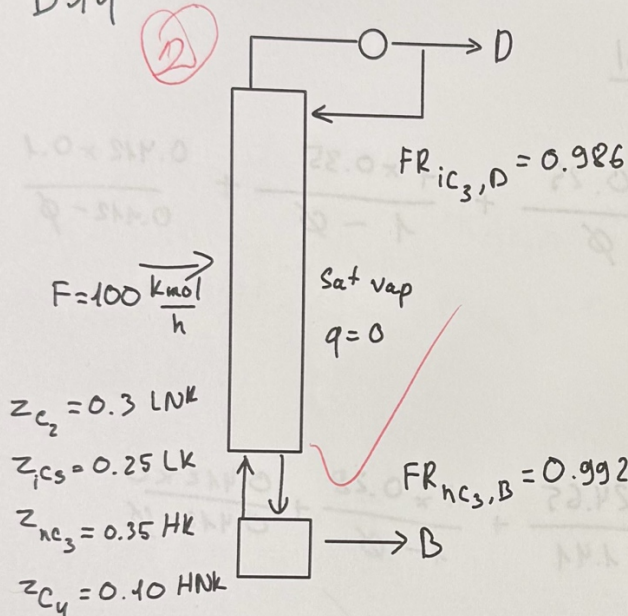
b) Same procedure

$$\phi = 0.3366$$

$$V_{\min} = 72.68 \frac{\text{kmol}}{\text{h}}$$

$$(L/D)_{\min} = 0.307$$

D14



Bottom

$$B x_{c_2, B} = 0$$

$$B x_{i c_3, B} = 0.35 \frac{\text{kmol}}{\text{h}}$$

$$B x_{n c_3, B} = 34.72 \frac{\text{kmol}}{\text{h}}$$

$$B x_{c_4, B} = 10 \frac{\text{kmol}}{\text{h}}$$

$$B = 45.07 \frac{\text{kmol}}{\text{h}}$$

$$x_{c_2, B} = 0$$

$$x_{i c_3, B} = 0.078$$

$$x_{n c_3, B} = 0.7703$$

$$x_{c_4, B} = 0.2219$$

Top

$$D x_{c_2, D} = 30 \frac{\text{kmol}}{\text{h}}$$

$$D x_{i c_3, D} = 24.65 \frac{\text{kmol}}{\text{h}}$$

$$D x_{n c_3, D} = 0.28 \frac{\text{kmol}}{\text{h}}$$

$$D x_{c_4, D} = 0$$

$$D = 54.93 \frac{\text{kmol}}{\text{h}}$$

$$x_{c_2, D} = 0.546$$

$$x_{i c_3, D} = 0.449$$

$$x_{n c_3, D} = 0.005$$

$$x_{c_4, D} = 0$$

$$b) N_{\min} = \frac{\ln \left(\frac{FR_{C3,D} \cdot FR_{C3,B}}{(1-FR_{C3,D})(1-FR_{C3,B})} \right)}{\ln 1.86} = 14.62 \text{ stages}$$

$$N_{F,\min} = \frac{\ln \left(\frac{x_{LK,D}}{x_{HK,D}} / \frac{z_{LK}}{z_{HK}} \right)}{\ln \alpha_{LK-HK}} = \frac{\ln \left(\frac{0.449}{0.005} / \frac{0.25}{0.35} \right)}{\ln 1.86} = 7.76 \text{ stages}$$

c) Same procedure

$$\Delta V_{\text{feed}} = F(1-q) = F = 100 \frac{\text{kmol}}{\text{h}}$$

$$1 = \frac{2.17 \times 0.3}{2.17 - \phi} + \frac{1.86 \times 0.25}{1.86 - \phi} + \frac{1 \times 0.35}{1 - \phi} + \frac{0.412 \times 0.1}{0.412 - \phi}$$

Solve for ϕ

$$\phi = 1.41$$

$$V_{\min} = \frac{2.17 \times 30}{2.17 - 1.41} + \frac{1.86 \times 24.65}{1.86 - 1.41} + \frac{1 \times 0.28}{1 - \phi} + \frac{0.412 \times 0}{0.412 - \phi}$$

$$= 186.86 \frac{\text{kmol}}{\text{h}}$$

$$\Rightarrow \left(\frac{L}{D} \right)_{\min} = \left(\frac{186.86 - 54.93}{54.93} \right)_{\min} = 2.40$$

$$\frac{L}{D} = 1.1 \left(\frac{L}{D} \right)_{\min} = 1.1 \times 2.40 = 2.64$$

$$\frac{L/D - (L/D)_{\min}}{L/D + 1} = 0.06593$$

$$\Rightarrow \frac{N - 14.62}{N + 1} = 0.545827 - 0.591422 \times 0.06593 + \frac{0.002713}{0.06593}$$

$$N = 33.59 \text{ stages}$$

$$\frac{N_{F,\min}}{N_{\min}} = \frac{N_F}{N} \Rightarrow N_F = 33.59 \times \frac{7.76}{14.62} = 17.83 \text{ stage} \approx 18 \text{ stage}$$